



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 2

2021 Year 12 Course Mock Assessment Task 1

Wednesday, 15 December 2021

General instructions

- Working time – 1 hour.
(plus 5 minutes reading time)
- Commence each new question on a new Writing Booklet. Write on both sides of the paper.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.
- Attempt all questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.
- A NESA Reference Sheet is provided.

NESA STUDENT #: # BOOKLETS USED:

Class: (please ✓)

- ☐ 12MXX.1 – Ms Ham
- ☐ 12MXX.2 – Mr Sekaran
- ☐ 12MXX.3 – Mr Sun

Marker's use only

QUESTION	1-8	9	10	11	Total	%
MARKS	$\overline{8}$	$\overline{10}$	$\overline{9}$	$\overline{11}$	$\overline{38}$	

Section I

8 marks Attempt Questions 1– 8

Allow about 10 minutes for this section

Use the multiple-choice answer sheet for Questions 1– 8.

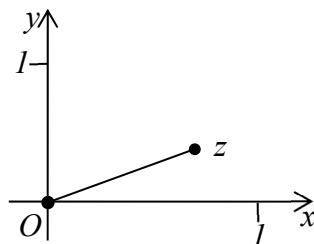
1. Let $z = a + ib$, where $a \neq 0$ and $b \neq 0$. Which statement is INCORRECT?

- (A) $z - \bar{z} = 2bi$ (B) $|z|^2 = |z||\bar{z}|$
 (C) $|z| + |\bar{z}| = |z + \bar{z}|$ (D) $\arg(z) + \arg(\bar{z}) = 0$

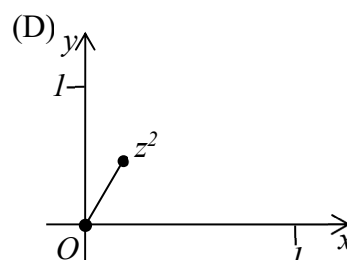
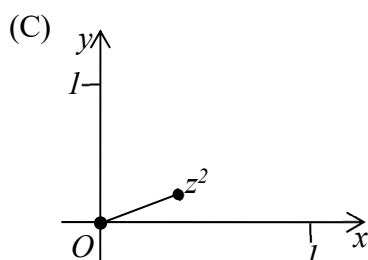
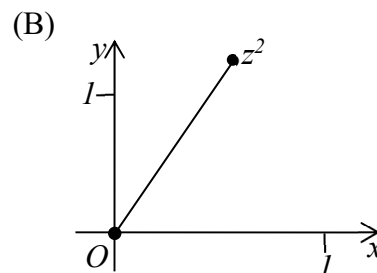
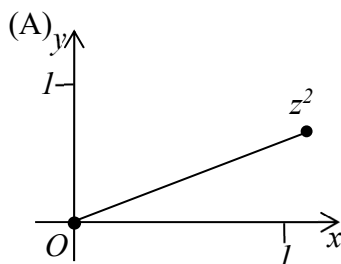
2. z satisfies the equation $|z - \sqrt{2} - i\sqrt{2}| = 1$. Find the minimum value of $\arg(z)$?

- (A) -75° (B) -15° (C) 15° (D) 75°

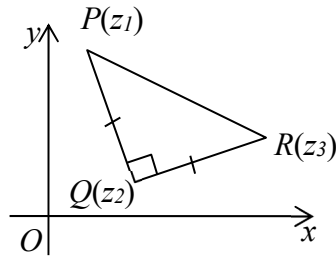
3. The Argand diagram below shows the complex number z .



Which diagram best represents z^2 ?

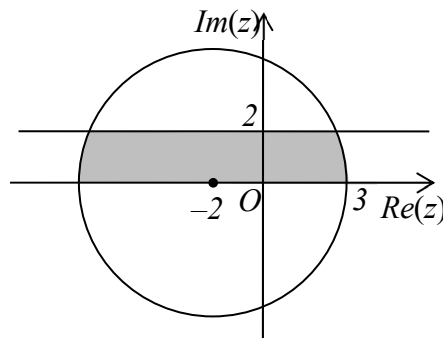


4. The vertices of the triangle PQR are represented by the complex numbers z_1, z_2 and z_3 respectively. The triangle PQR is isosceles and right-angled at Q , as shown in the diagram.



Which of the following statement is true?

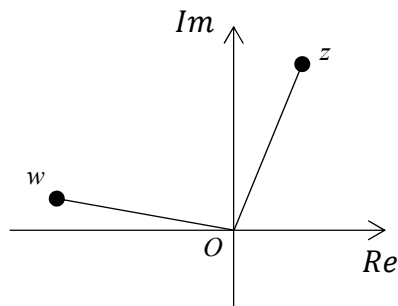
- (A) $z_2 - z_1 = i(z_3 - z_2)$ (B) $z_1 - z_2 = i(z_3 - z_2)$
 (C) $z_2 - z_1 = i(z_1 - z_3)$ (D) $z_1 - z_2 = i(z_1 - z_3)$
5. Let the cube roots of unity be $1, \omega, \omega^2$. The value of $(1 - \omega - \omega^2)(1 + 2\omega + 2\omega^2)$ is
 (A) 0 (B) ω (C) 2 (D) -2
6. A circle with centre $(-2, 0)$ and radius 5 units is shown on an argand diagram below.



Which set of the following inequalities represents the shaded region?

- (A) $Re(z) \leq 0, Im(z) \leq 2$ and $|z - 2| \leq 3$ (B) $Im(z) \geq 0, Im(z) \leq 2$ and $|z - 2| \geq 3$
 (C) $Im(z) \geq 0, Im(z) \leq 2$ and $|z - 2| \leq 5$ (D) $Re(z) \leq 2, Im(z) \geq 0$ and $|z - 2| \leq 5$

7. The Argand diagram shows the complex numbers z and w , where z lies in the first quadrant and w lies in the second quadrant.



Which complex number could lie in the 3rd quadrant?

- (A) $-w$ (B) $2iz$ (C) $\bar{z}$ (D) $w - z$
8. It is given that $z = 2 + i$ is a root of $z^3 + az^2 - 7z + 15 = 0$, where a is a real number.
- What is the value of a ?
- (A) -1 (B) 1 (C) 7 (D) -7

Section II 31 marks

Attempt Questions 9 - 11

Allow about 50 minutes for this section

Answer each question in the appropriate writing booklet.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 9	(10 marks)	Marks
(a)	Find the complex square roots of $7 + 6\sqrt{2}i$ giving your answers in the form $x + iy$, where x and y are real.	3
(b)	If $z = \frac{1}{\sqrt{2}}(1 - i)$, find z^{2022} in the form $a + ib$, where a and b are real.	2
(c)	i. On the same diagram, sketch the region of the complex plane defined by each of the following: $\alpha.$ $ z - (3 + 2i) = 2$ $\beta.$ $ z + 3 = z - 5 $	1 1
	ii. Hence write down all values of z which satisfy simultaneously $ z - (3 + 2i) = 2$ and $ z + 3 = z - 5 $.	1
	iii. Use your diagram in (i) to determine the values of k for which the simultaneous equations $ z - (3 + 2i) = 2$ and $ z - 2i = k$ have exactly one solution for z .	2

Question 10 (9 marks)

- (a) It is known that $2 + 5i$ is a zero of the polynomial $P(z) = z^3 - 8z^2 + 45z - 116$.
- Find the third zero of $P(z)$. 2
 - Hence write $P(z)$ as a product of two factors with real coefficients. 2
- (b) Let $w = \cos \frac{2\pi}{9} + i \sin \frac{2\pi}{9}$.
- Show that w^k is a solution of $z^9 - 1 = 0$, where k is an integer. 1
 - Prove that $w + w^2 + w^3 + w^4 + w^5 + w^6 + w^7 + w^8 = -1$. 1
 - Write $w^9 - 1$ as a product of one linear and four quadratic factors, all with real coefficients. 1
 - Hence or otherwise, show that $\cos\left(\frac{\pi}{9}\right) \cos\left(\frac{2\pi}{9}\right) \cos\left(\frac{4\pi}{9}\right) = \frac{1}{8}$. 2

Question 11 (11 marks)

- (a) Suppose that $z = 4\sqrt{3}e^{\frac{i\pi}{3}} - 4e^{\frac{i5\pi}{6}}$.
- Simplify z , writing your answer in exponential form. 2
 - Show that $\left(\frac{z}{8}\right)^3 + i\left(\frac{z}{8}\right)^2 + \frac{z}{8} = 2i$, clearly show all working. 2
 - Find the three cube roots of z in exponential form. 2
- (b) Let the points $A_1, A_2, \dots, A_n$ represent the n th roots of unity, $w_1, w_2, \dots, w_n$, and suppose P represents any complex number z such that $|z| = 1$.
- Prove that $w_1 + w_2 + \dots + w_n = 0$. 1
 - Show that $PA_i^2 = (z - w_i)(\bar{z} - \bar{w}_i)$ for $i = 1, 2, \dots, n$. 1
 - Prove that $\sum_{i=1}^n PA_i^2 = 2n$. 3

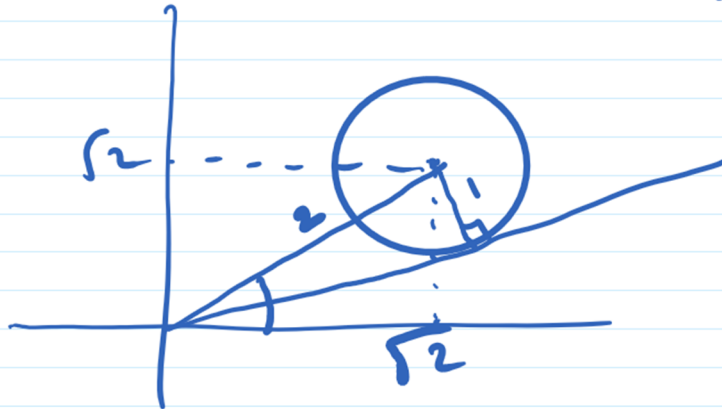
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Sample solutions mock assessment task 1-2022

Q1 C

Q2 C $|z - \sqrt{2} - \sqrt{2}i| = 1$

$$4 + 2 = \sqrt{6}$$



$$\begin{aligned} \tan^{-1}\left(\frac{\sqrt{2}}{\sqrt{2}}\right) - \sin^{-1}\left(\frac{1}{2}\right) \\ = 45^\circ - 30^\circ \\ = 15^\circ \end{aligned}$$

Q3 D

Q4 B

Q5 D $(1 - \omega - \omega^2)(1 + 2\omega + 2\omega^2)$

$$1 + \omega + \omega^2 = 0$$

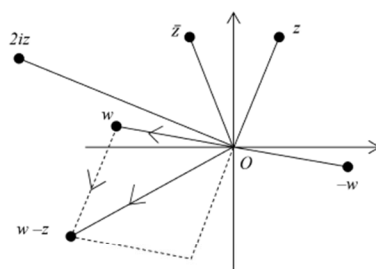
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$$\begin{aligned} &= (1 + 1)(1 + 2(-1)) \\ &= -2 \end{aligned}$$

Q6 C

Q7 D

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Q8 A

$$z = 2 + i, z^2 = 4 + 4i + i^2 = 3 + 4i,$$

$$z^2(z + a) - 7z + 15 = 0,$$

$$(3 + 4i)(2 + i + a) - 7(2 + i) + 15 = 0$$

$$\Rightarrow 6 + 3i + 3a + 8i - 4 + 4ai - 14 - 7i + 15 = 0$$

$$\Rightarrow 3a + 4ai = -3 - 4i \Rightarrow a = -1 \therefore A$$

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Q9

a) $(x+iy)^2 = 7+6\sqrt{2}i$, $x^2 - y^2 = 7$ and $2xy = 6\sqrt{2}$

i) $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2 = 7^2 + (6\sqrt{2})^2 = 121$

$x^2 + y^2 = 11 (x^2 + y^2 \geq 0)$, $2x^2 = 18$, $x = \pm 3$, $y = \pm\sqrt{2}$

So the two square roots are $\pm(3+i\sqrt{2})$

b) $z = \frac{1}{\sqrt{2}}(1-i) = e^{-i\frac{\pi}{4}}$

$504 + \frac{\pi}{2}$

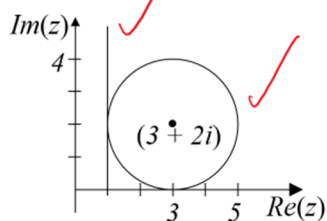
$z^{2022} = e^{-i\frac{1011}{2}\pi}$

$= 1 \cdot e^{-i\frac{2\pi}{2}} = -i(-1) = i = 0+i$

c)

α) The locus is a circle centre $3+2i$ and radius 2.

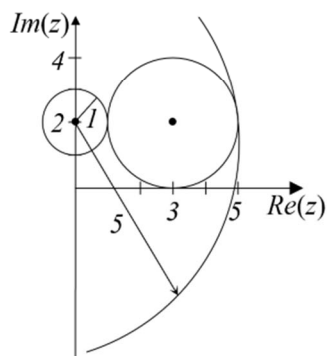
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β) the locus is the points equidistant from -3 and 5 . i.e. $x = 1$

ii) On one point: $z = 1 + 2i$

iii) $|z - 2i| = k$ represents a circle, centre $2i$ and radius k . This circle touches the first circle if $k = 1$ or $k = 5$



Question 10 (8 marks)

(a) It is known that $2 + 5i$ is a zero of the polynomial $P(z) = z^3 - 8z^2 + 45z - 116$.

i. Find the third zero of $P(z)$.

2

ii. Hence write $P(z)$ as a product of two factors with real coefficients.

2

i) zeros are $2 + 5i, 2 - 5i, \checkmark$
 $\therefore 2 + 5i + 2 - 5i + \alpha = 8 \checkmark$
 $\alpha = 4 \checkmark$

$\therefore$ third zero is 4

ii) $P(z) = (z - (2 + 5i))(z - (2 - 5i))(z - 4) \checkmark$
 $= (z^2 - (2 + 5i + 2 - 5i)z + (2 + 5i)(2 - 5i))(z - 4)$
 $= (z^2 - 4z + 29)(z - 4) \checkmark$

b)

i) $w^k = \cos \frac{2\pi k}{9} + i \sin \frac{2\pi k}{9}$

$\therefore z^9 - 1 = (w^k)^9 - 1$

$= (\cos 2\pi k + i \sin 2\pi k) - 1 = (1 + 0) - 1 = 0 \checkmark$

If k is an integer, w^k is a solution to $z^9 - 1 = 0$

ii) Using the sum of a G.P $w + w^2 + w^3 + \dots + w^8$

$= \frac{w(1 - w^9)}{1 - w} = \frac{w - w^9}{1 - w} = \frac{w - 1}{1 - w} = -1 \checkmark$

iii) roots are $w_1 = 1$
 $w_2 = \cos \frac{2\pi}{9} + i \sin \frac{2\pi}{9}$
 $w_3 = \overline{w_2}$
 $w_4 = \cos \frac{4\pi}{9} + i \sin \frac{4\pi}{9}$
 $w_5 = \overline{w_4}$
 $w_6 = \cos \frac{6\pi}{9} + i \sin \frac{6\pi}{9}$
 $w_7 = \overline{w_6}$
 $w_8 = \cos \frac{8\pi}{9} + i \sin \frac{8\pi}{9}$
 $w_9 = \overline{w_8}$

$w^9 - 1 = (w - 1)(w - \cos \frac{2\pi}{9} - i \sin \frac{2\pi}{9})(w - \cos \frac{2\pi}{9} + i \sin \frac{2\pi}{9})$
 $(w - \cos \frac{4\pi}{9} - i \sin \frac{4\pi}{9})(w - \cos \frac{4\pi}{9} + i \sin \frac{4\pi}{9})$
 $(w - \cos \frac{6\pi}{9} - i \sin \frac{6\pi}{9})(w - \cos \frac{6\pi}{9} + i \sin \frac{6\pi}{9})$
 $(w - \cos \frac{8\pi}{9} - i \sin \frac{8\pi}{9})(w - \cos \frac{8\pi}{9} + i \sin \frac{8\pi}{9})$
 $= (w - 1)(w^2 - 2 \cos \frac{2\pi}{9} w + 1)(w^2 - 2 \cos \frac{4\pi}{9} w + 1)(w^2 - 2 \cos \frac{6\pi}{9} w + 1)(w^2 - 2 \cos \frac{8\pi}{9} w + 1)$

iv) Sub $w = i$
 $i^9 - 1 = (i - 1)(-2 \cos \frac{2\pi}{9} i)(-2 \cos \frac{4\pi}{9} i)(-2 \cos \frac{6\pi}{9} i)(-2 \cos \frac{8\pi}{9} i)$

$\therefore 1 = (-2)^4 \cos \frac{2\pi}{9} \cos \frac{4\pi}{9} \cos \frac{6\pi}{9} \cos \frac{8\pi}{9}$
 $\therefore \cos \frac{2\pi}{9} \cos \frac{4\pi}{9} \cos \frac{6\pi}{9} \cos \frac{8\pi}{9} = \frac{1}{8} \checkmark$

Alternate method

$$\therefore \cos \frac{2\pi}{9} \cos \frac{4\pi}{9} \cos \frac{8\pi}{9} = \frac{1}{8}$$

10) ~~iii)~~ $z^9 - 1 = 0$ has 9 distinct roots: $w^k = \cos \frac{2\pi k}{9} + i \sin \frac{2\pi k}{9}$, where $k = 0, 1, 2, \dots, 9$

From (ii): $(w + w^8) + (w^2 + w^7) + (w^3 + w^6) + (w^4 + w^5) = -1$

Note that in each bracket there is a pair of conjugate complex numbers.

$$\text{For example: } w + w^8 = \cos \frac{2\pi}{9} + i \sin \frac{2\pi}{9} + \cos \frac{16\pi}{9} + i \sin \frac{16\pi}{9}$$

$$= \cos \frac{2\pi}{9} + i \sin \frac{2\pi}{9} + \cos \left(-\frac{2\pi}{9} \right) + i \sin \left(-\frac{2\pi}{9} \right) = 2 \cos \frac{2\pi}{9}$$

The other brackets yield similar results.

$$2 \cos \frac{2\pi}{9} + 2 \cos \frac{4\pi}{9} + 2 \cos \frac{6\pi}{9} + 2 \cos \frac{8\pi}{9} = -1$$

$$2 \left(\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} \right) + 2 \left(\cos \frac{6\pi}{9} + \cos \frac{8\pi}{9} \right) = -1$$

$$4 \cos \frac{\pi}{9} \cos \frac{3\pi}{9} + 4 \cos \frac{\pi}{9} \cos \frac{7\pi}{9} = -1$$

$$4 \cos \frac{\pi}{9} \left(\cos \frac{3\pi}{9} + \cos \frac{7\pi}{9} \right) = -1$$

$$8 \cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9} = -1$$

$$-\cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9} = -\frac{1}{8}$$

$$\Rightarrow \cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9} = \frac{1}{8}$$

Question 11

z

$$\begin{aligned} &= 4\sqrt{3}e^{\frac{i\pi}{3}} - 4e^{\frac{5i\pi}{6}} \\ &= 4\sqrt{3}\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) - 4\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right) \\ &= 4\sqrt{3}\left(\frac{1}{2} + \frac{i\sqrt{3}}{2}\right) - 4\left(-\frac{\sqrt{3}}{2} + \frac{i}{2}\right) \\ &= 4\sqrt{3} + 4i \\ &= 8\left(\frac{\sqrt{3}}{2} + \frac{i}{2}\right) \\ &= 8e^{\frac{i\pi}{6}} \end{aligned}$$

Using part a we have

$$\begin{aligned} &\frac{z}{8} + i\left(\frac{z}{8}\right)^2 + \left(\frac{z}{8}\right)^3 \\ &= \frac{8e^{\frac{i\pi}{6}}}{8} + i\left(\frac{8e^{\frac{i\pi}{6}}}{8}\right)^2 + \left(\frac{8e^{\frac{i\pi}{6}}}{8}\right)^3 \\ &= e^{\frac{i\pi}{6}} + i\left(e^{\frac{i\pi}{6}}\right)^2 + \left(e^{\frac{i\pi}{6}}\right)^3 \\ &= e^{\frac{i\pi}{6}} + ie^{\frac{i\pi}{3}} + e^{\frac{i\pi}{2}} \\ &= \cos\frac{\pi}{6} + i\sin\frac{\pi}{6} + i\cos\frac{\pi}{3} + i^2\sin\frac{\pi}{3} + \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} \\ &= \frac{\sqrt{3}}{2} + \frac{i}{2} + \frac{i}{2} - \frac{\sqrt{3}}{2}i + 0 + i \\ &= 2i \end{aligned}$$

Let $\lambda = re^{i\theta}$ be a cube root of z . It follows that

$$\lambda^3 = r^3e^{3i\theta} = 8e^{\frac{i\pi}{6}}$$

Comparing modulus, we see that $r^3 = 8$ and so $r = 2$. Then comparing argument, we see that $3\theta = 2n\pi + \frac{\pi}{6} = \frac{(12n+1)\pi}{6}$, where n is an integer. Hence,

$$\theta = \frac{(12n+1)\pi}{18}$$

Thus, the roots are of the form

$$\lambda = 2e^{\frac{(12n+1)\pi}{18}} \text{ where } n \text{ is an integer}$$

Taking $n = -1, 0, 1$, we see that the three cube roots of z are,

$$\lambda = 2e^{-\frac{11i\pi}{18}}, 2e^{\frac{i\pi}{18}}, 2e^{\frac{13i\pi}{18}}$$

$$\begin{aligned} \text{ii) } PA_i &= |z - w_i|, PA_i^2 = |z - w_i|^2 \\ &= (z - w_i)(\overline{z - w_i}) = (z - w_i)(\bar{z} - \bar{w}_i) \end{aligned}$$

$$\begin{aligned} PA_i^2 &= (w_i - \bar{z})(\bar{w}_i - \bar{z}) \\ &= w_i \bar{w}_i - w_i \bar{z} - \bar{w}_i \bar{z} + \bar{z} \bar{z} \\ &= |w_i|^2 - w_i \bar{z} - \bar{w}_i \bar{z} + |\bar{z}|^2 \\ &= 2 - w_i \bar{z} - \bar{w}_i \bar{z} \end{aligned}$$

$$\sum_{i=1}^n p A_i^2 = \sum_{i=1}^n (2 - \omega_i \bar{z} - \bar{\omega}_i z)$$

$$= \sum_{i=1}^n z^2 - \sum_{i=1}^n \omega_i \bar{z} - \sum_{i=1}^n \bar{\omega}_i z$$

$$= 2n - \bar{z} \sum_{i=1}^n w_i - z \sum_{i=1}^n \bar{w}_i$$

$$= 2n - \bar{z} \cdot 0 - z (\overline{w_1 + w_2 + \dots + w_n})$$

$$\sum_{i=1}^n 2 = 2 + 2 + \dots + 2 = 2n$$